

Module 2

Chapter 2

More about Vectors and Matrices

Some special Matrices

We already know by now that Matrices are a group of elements arranged in rows and columns. In general, an $m \times n$ matrix is a rectangular array of $m*n$ numbers (or elements) arranged in m rows and n columns. If $m = n$ the matrix is called a **square matrix**.

Here's how a 2*2 matrix will look like:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Generally, we use bold phase letter, like **A**, to denote a matrix, and lower-case letters with subscripts, like a_{ij} , to denote element of a matrix. Here a_{ij} would be the element at **i^{th} row** and **j^{th} column**. So a_{11} is an element at 1st row and column.

Zero Matrix:

0 denotes the zero matrix whose elements are all *zeroes*. So, a 2*2 zero matrix would look like:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Identity Matrix:

An identity matrix is an matrix whose main diagonal elements are 1, and all other elements are 0. So, a 3*3 Identity Matrix will look like:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Vectors:

A matrix with a single row or column is called a vector. Vectors with a single column are called *Column Vectors* and the ones with a single row are called *Row Vectors*.

Vectors are usually represented as lower case letters with an arrow above them, like, $\vec{a}$. A 3D row vector will look like:

$$[x_1 \quad x_2 \quad x_3]$$

While a 3D column vector will look like:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Operations on Vectors and Matrices:

- **Equality:** Two matrix $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ are equal if corresponding elements are equal, i.e. $a_{ij} = b_{ij}$.
- **Addition:** If $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$, then the sum of $\mathbf{A}$ and $\mathbf{B}$ is $\mathbf{C} = \mathbf{A} + \mathbf{B} = (c_{ij}) = a_{ij} + b_{ij}$.
- **Scalar Product:** If $\mathbf{A} = (a_{ij})$ is matrix and k is number (scalar), then $k\mathbf{A} = (ka_{ij})$ is product of k and $\mathbf{A}$.

Similarly, applying the above operations, we have: $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-1)\mathbf{B}$.

Since vectors are special matrices, the same rules apply to vectors as well!

Let's Practice:

Let $\mathbf{A}$ be:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

and $\mathbf{B}$ be:

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 4 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Find the following:

1. $\mathbf{A} + \mathbf{B}$
2. $4\mathbf{A}$
3. $3\mathbf{A} - 4\mathbf{B}$

$$A+B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 2 & 4 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 3 \\ 6 & 6 & 2 \\ 1 & 0 & 2 \end{bmatrix}$$

$$4A = \begin{bmatrix} 4 & 8 & 12 \\ 16 & 8 & 4 \\ 0 & 0 & 4 \end{bmatrix}$$

$$3A - 4B = \begin{bmatrix} 3 & 6 & 9 \\ 12 & 6 & 3 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 8 & 16 & 4 \\ 4 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 6 & 9 \\ 4 & -10 & -1 \\ -4 & 0 & -1 \end{bmatrix}$$

The following fact lists all properties of matrix addition and scalar product:

Let $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ be matrices. Let a , b be scalars (numbers). We have

- $\mathbf{A} + \mathbf{0} = \mathbf{0} + \mathbf{A} = \mathbf{A}$, $\mathbf{A} - \mathbf{A} = \mathbf{0}$;
- $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ (commutativity);
- $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$, $(ab)\mathbf{A} = a(b\mathbf{A})$ (associativity);
- $a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}$, $(a+b)\mathbf{A} = a\mathbf{A} + b\mathbf{A}$ (distributivity).

Length of a vector:

Length of a vector, also called its magnitude, can be calculated by taking the square root of the sum of the squares of its components. It is usually, but not always, considered to be an extension of the Pythagorean theorem:

$$|\vec{v}| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

A vector with magnitude 1 is called a Unit Vector

Let's Practice:

Let

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ and } \vec{b} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

Find:

- $3\vec{a} + 3\vec{b}$
- $|\vec{a}|$
- $|3\vec{a} + 2\vec{b}|$
- $|3\vec{a}| + |2\vec{b}|$

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$3\vec{a} + 3\vec{b} = 3(\vec{a} + \vec{b}) = 3 \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 12 \\ 12 \\ 12 \end{bmatrix}$$

$$|\vec{a}| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$$

$$|3\vec{a} + 2\vec{b}| = \left| \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix} \right| = \left| \begin{bmatrix} 9 \\ 10 \\ 11 \end{bmatrix} \right|$$

$$= \sqrt{9^2 + 10^2 + 11^2} = \sqrt{81 + 100 + 121}$$

$$= \sqrt{302} \approx 17.38$$

$$|3\vec{a}| + |2\vec{b}| = \sqrt{9 + 36 + 81} + \sqrt{36 + 16 + 4}$$

$$= \sqrt{126} + \sqrt{56}$$

$$\approx 11.22 + 7.5 = 18.72$$

Dot Product of two vectors:

A very common operation to combine two vectors is the dot product. When we have two vectors with the same number of elements, we can define the dot product as

$$\vec{a} \cdot \vec{b} = a_1 \cdot b_1 + a_2 \cdot b_2 \dots + a_n \cdot b_n$$

For example, if

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ and } \vec{b} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

Then,

$$\vec{a} \cdot \vec{b} = 1 \cdot 3 + 2 \cdot 2 + 3 \cdot 1 = 10$$

Remember, the result of dot product of two vectors is always a scalar quantity!!

There is another formula for dot product of two vectors which is especially used to find the angle between two vectors:

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

Here, θ is the angle between $\vec{a}$ and $\vec{b}$

$a_n b_n$

So, if

$\vec{a} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, then their dot product is 0 using

$$\vec{a} \cdot \vec{b} = a_1 \cdot b_1 + a_2 \cdot b_2 + \dots + a_n \cdot b_n$$

But, we also know that,

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

This implies that:

$$0 = |\vec{a}| |\vec{b}| \cos \theta$$

We know that

$$|\vec{a}| \text{ and } |\vec{b}| \neq 0, \text{ thus, } \cos \theta = 0$$

This gives us $\theta = 90^\circ$. Vectors which have an angle of 90° between them are called **Orthogonal** vectors. / perpendicular

Matrix Multiplication:

Let $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$, if the number of columns of $\mathbf{A}$ is the same as number of rows of $\mathbf{B}$, then the product of $\mathbf{A}$ and $\mathbf{B}$ is given by $\mathbf{AB} = (c_{ij})$ where c_{ij} is dot product of i^{th} row of $\mathbf{A}$ with j^{th} column of $\mathbf{B}$.

Remember $\mathbf{AB}$ is valid if and only if the number of columns of $\mathbf{A}$ is the same as number of rows of $\mathbf{B}$

So, if

$$\mathbf{A} = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 0 & 5 \\ 3 & -4 \end{bmatrix} \text{ then}$$

$$\mathbf{AB} = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 0 & 5 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} 2.(0) + 3.(3) & 2.(5) + 3.(-4) \\ -1.(0) + 4.(3) & -1.(5) + 4.(-4) \end{bmatrix} = \begin{bmatrix} 9 & -2 \\ 12 & -21 \end{bmatrix}$$

Let's Practice:

- Find the value of α for which vector $[5, -2\alpha, 2]$ is orthogonal to the vector $[1, -1]$.
- Find all two-dimensional vectors $\vec{a}$ orthogonal to vector $\vec{b} = [3, 4]$
- For $\vec{a} = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$, calculate the cosine of the angle between the two vectors.
- Let $M = \begin{bmatrix} 2 & 8 & 3 \\ 5 & 4 & 1 \end{bmatrix}$ and $N = \begin{bmatrix} 4 & 1 \\ 6 & 3 \\ 2 & 4 \end{bmatrix}$, find MN

1. Since the vectors are orthogonal, $\theta = 90^\circ \therefore \cos \theta = 0$

$$\therefore |a||b| \cos \theta = 0 = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\vec{a} = (5, -2d, 2) \quad \vec{b} = (1, -1) \quad \text{--- NOT the same dimensions}$$

Here the dimensions of $\vec{b}$ are not the same as dimensions of $\vec{a}$.

$$\therefore \vec{b} \text{ can be } \rightarrow \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Taking each case individually,

$$\text{i)} \quad \begin{bmatrix} 5 \\ -2d \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = 0$$

$$\Rightarrow 5 - 2d + 0 = 0$$

$$\Rightarrow d = -5/2$$

$$\text{ii)} \quad \begin{bmatrix} 5 \\ -2d \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = 0$$

$$\Rightarrow -2d - 2 = 0$$

$$\Rightarrow d = 1$$

$$\text{iii)} \quad \begin{bmatrix} 5 \\ -2d \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = 0$$

$$\Rightarrow 5 - 2 = 0 \rightarrow \text{Not possible}$$

To avoid confusion,
the dimensions of
a vector are often
written using unit
vectors: $\hat{i}, \hat{j}, \hat{k}$

$\hat{i} \rightarrow$ unit vector in
x-direction

$\hat{j} \rightarrow$ unit vector in
y-direction

$\hat{k} \rightarrow$ unit vector in
z-direction

2. $\vec{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

Let $\vec{a}$ be $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$

$\therefore 3a_1 + 4a_2 = 0$

$\Rightarrow a_2 = -\frac{3a_1}{4}$

$\therefore$ all vectors of the form $\begin{bmatrix} a_1 \\ -\frac{3a_1}{4} \end{bmatrix}$ are orthogonal to $\vec{b}$

Eg. $\begin{bmatrix} 4 \\ -3 \end{bmatrix}$ or $\begin{bmatrix} 8 \\ -6 \end{bmatrix}$ and so on

3. $\vec{a} = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$ $\vec{b} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$

$\vec{a} \cdot \vec{b} = (\sqrt{4+1+4}) (\sqrt{9+4+1}) \cos \theta$

$\vec{a} \cdot \vec{b} = 6 - 2 + 2 = 6$

$\therefore \sqrt{9} \cdot \sqrt{14} \cos \theta = 6$

$\therefore \cos \theta = \frac{6}{3\sqrt{14}} = \frac{2\sqrt{14}}{14} = \frac{\sqrt{14}}{7}$

$$3. \quad M = \begin{bmatrix} 2 & 8 & 3 \\ 5 & 4 & 1 \end{bmatrix} \quad N = \begin{bmatrix} 4 & 1 \\ 6 & 3 \\ 2 & 4 \end{bmatrix}$$

$2 \times 3 \qquad \qquad \qquad 3 \times 2$

$\therefore$ Valid

$$MN = \begin{bmatrix} 8 + 48 + 6 & 2 + 24 + 12 \\ 20 + 24 + 2 & 5 + 12 + 4 \end{bmatrix}$$

2×2

$$MN = \begin{bmatrix} 62 & 38 \\ 46 & 21 \end{bmatrix}$$